

DIFFERENCE BETWEEN DATA STRUCTURE AND ADVANCEDDATA STRUCTURES & ALGORITHM ANALYSIS

| ASPECT         | DATA STRUCTURES                                                                                   | ADSAA                                                                                                                                                    |
|----------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEFINITION     | Study of organizing and storing data efficiently                                                  | Study of advanced data structures and analysis of algorithms for efficient problem-solving                                                               |
| Main focus     | Data organization and implementation                                                              | Advanced data structuring, Algorithm design and performance analysis.                                                                                    |
| Objective      | Learn how to store, access & manage data efficiently                                              | Learn how to design efficient algorithms and analyze their time & space requirements                                                                     |
| TOPICS covered | Arrays, Linked lists, Stacks, Queues, Trees, Heaps, Hash tables, Basic Graphs, Searching, Sorting | AVL Trees, Red Black trees, B-Trees, Advanced Graphs, algorithms, Dynamic Programming, Greedy Algorithms, Divide & Conquer, Backtracking, Branch & Bound |

|                    |                                                      |                                                                                                          |
|--------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Algorithm Analysis | Basic Introduction to searching & sorting algorithms | Detailed analysis using Big O, Big Theta, Big Omega, recurrence relations & algorithm Optimization.      |
| Complexity         | Covers basic time & space complexity                 | focuses extensively on algorithm complexity & efficiency.                                                |
| Difficulty level   | Beginner to Intermediate                             | Intermediate to Advanced                                                                                 |
| Prerequisite       | Basic programming knowledge                          | Strong understanding of data structures & programming concepts                                           |
| Applications       | Data storage, databases OS & s/w development         | AI, n/w, Optimization, Competitive programming & large-scale s/w Systems.                                |
| Learning outcome   | Ability to implement & use standard data             | Ability to design efficient algorithm, analyze their performance & solve complex computational problems. |

# Application Real time for ADSSAA

## Algorithm used

## Purpose

1. Google maps — Graph Algorithm — (Dijkstra's) — find the shortest and fastest route b/w locations
2. GPS Navigation — Graphs, Greedy Algorithms — Suggest the best route based on traffic & distance
3. Social media (Instagram/ Facebook) — Graphs — Suggest friends & connections
4. Search Engine — B-Trees, Hash Tables — Quickly search & retrieve web pages
5. E-commerce (Amazon, Flipkart) — Sorting, Searching Hash Tables. — Displays products based on price, rating & relevance
6. Banking System — AVL Trees, B Trees. — fast storage & retrieval of customer records
7. Ride sharing (Uber, Ola) — Graph Algorithms — Match drivers to passengers & calculate optimal routes
8. Video Streaming (YouTube, Netflix) — Dynamic Programming, — Recommended Videos & movies

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|---------------------|-------------------------------------|--------------------------------------------------|
| 9. Online Game      | Graph Algorithms,<br>Priority Queue | - Find Shortest Paths & manages Player Ranking   |
| 10. Compiler Design | Trees, Graphs                       | - Optimizes & translates Source code efficiently |

## DS

- |                | <u>Realtime</u><br><u>Ex</u>      | <u>Purpose</u>                              |
|----------------|-----------------------------------|---------------------------------------------|
| 1. Array       | Student mark list                 | Stores data in a fixed sequence             |
| 2. Linked list | Music Playlist                    | Allows easy insertion and deletion of songs |
| 3. Stack       | Browser back button               | LIFO                                        |
| 4. Queue       | Printer Queue                     | FIFO                                        |
| 5. Tree        | file & folder system              | Organizes data hierarchically               |
| 6. BST         | Dictionary (or)<br>Phone contacts | Enable faster searching of records          |
| 7. Heap        | Hospital emergency queue          | Gives priority to the most urgent cases     |
| 8. Hash table  | Login system                      |                                             |
| 9. Graph       | Social media friend N/w           | Represents relationship between users.      |